Neap

HSC Trial Examination 2020

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–29)

- Attempt Questions 11–28
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks
Attempt Questions 1–10
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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1. What is the value of $\operatorname{cosec}\frac{\pi}{3}$ correct to three significant figures?

(A) 1.00

(B) 1.15

(C) 1.41

(D) 2.00

2. What is the amplitude of $f(x) = -3\sin\left(2x + \frac{\pi}{3}\right)$?

(A) $-\pi$

(B) -3

(C) 3

(D) π

3. What is the natural domain of $f(x) = \frac{1}{e^x}$?

(A) $(-\infty, \infty)$

(B) $[0, \infty)$

(C) $(0, \infty)$

(D) $(-\infty, 0]$

4. What is the equation of the tangent to $y = x^2 - 3$ at $x = -1$?

(A) $y = -2x - 4$

(B) $y = 2x - 4$

(C) $y = \frac{x}{2} - \frac{3}{2}$

(D) $y = -\frac{x}{2} - \frac{3}{2}$

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5. Which one of the following is the set of all solutions to $2x^2 - 5x + 2 \geq 0$?

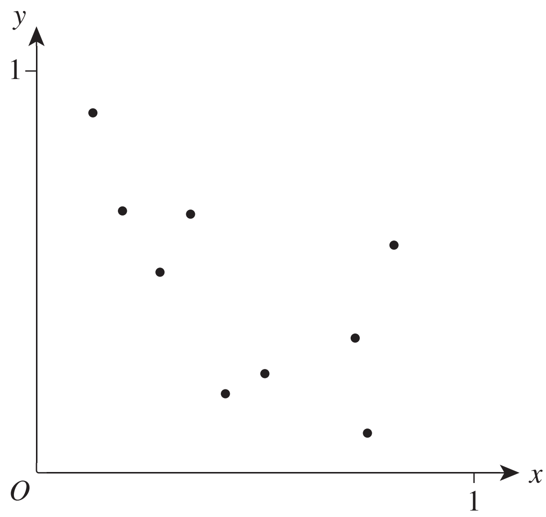
(A) $\left[\frac{1}{2}, 2\right]$

(B) $\left(\frac{1}{2}, 2\right)$

(C) $\left(-\infty, \frac{1}{2}\right) \cup (2, \infty)$

(D) $\left(-\infty, \frac{1}{2}\right] \cup [2, \infty)$

6. The scatter plot relates the quantities x and y.



Which one of the following values is the most appropriate Pearson's correlation coefficient for x and y?

(A) -0.97

(B) -0.63

(C) 0.12

(D) 0.55

7. The graph of $y = f(x)$ has a stationary point at $(2, -3)$.

Which one of the following is a guaranteed stationary point of $y = -f\left(\frac{x}{2}\right) - 5$?

(A) $(1, -2)$

(B) $(1, 2)$

(C) $(4, -2)$

(D) $(4, 2)$

8. Which one of the following equations is NOT correct?

(A) $\displaystyle\int x(x^2-1)^2\,dx = \frac{(x^2-1)^3}{6} + c$

(B) $\displaystyle\int_{-3}^{3} \sqrt{9-x^2}\,dx = \frac{9\pi}{2}$

(C) $\displaystyle\int_{-1}^{1} 3^x\,dx = \frac{1}{\ln 3}\left(3-\frac{1}{3}\right)$

(D) $\displaystyle\int_{-5}^{5} 4x^4 - x^3 + \cos x\,dx = 0$

9. A particle moves according to the equation $x = t^2 - \ln(t+1)$, where $t \geq 0$.

How may times does the particle come to rest?

(A) 0
(B) 1
(C) 2
(D) 3

10. An endurance test requires participants to consistently jog around a 400 m track. A participant passes the test if they successfully jog 1 full lap. A participant is said to be ‘Very Fit’ if they successfully jog 3 full laps.

Where X is the number of full laps a participant successfully jogs, the distribution function of X is $P(X = x) = p(x) = 0.2(0.8)^k$ for $k = 0, 1, 2, \ldots$

What is the probability that a participant who has passed the test is Very Fit; that is, what is $P(X \geq 3 | X \geq 1)$?

(A) 0.288
(B) 0.512
(C) 0.64
(D) 0.8

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Section II

90 marks
Attempt Questions 11–28
Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

What angle does the line $-x + 4y + 12 = 0$ make with the positive x-axis? Round to the nearest minute. **2**

Question 12 (3 marks)

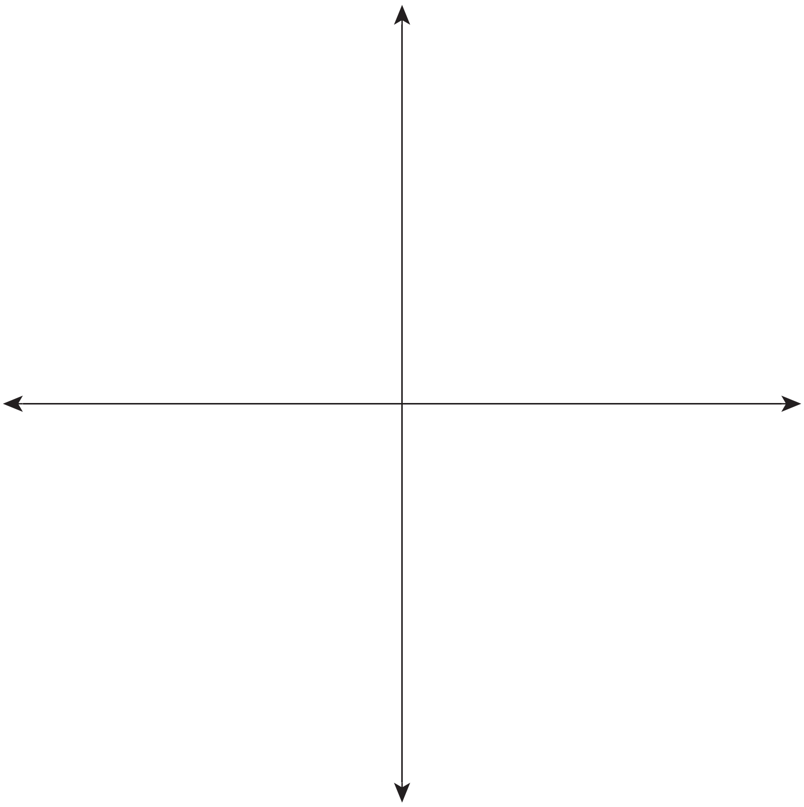
Differentiate the following expressions.

(a) $\log_2(\cos x)$ **1**

(b) $3^x e^x$ **2**

Question 13 (3 marks)

Sketch $y = \dfrac{3}{x+2} + 2$ on the axes below, showing all intercepts with the coordinate axes **3**
and all asymptotes. Show your working.



Question 14 (3 marks)

Eye colour is recorded in a population of 266 male and female subjects.

(a) Complete the two-way table for the data collected. **2**

	Male	*Female*	*Total*
Brown	98		190
Other		37	76
Total	137		

(b) A random person is chosen from this population. **1**

Given that this person's eye colour is brown, what is the probability that they are female?

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Question 15 (2 marks)

An amount of money is invested in an account that earns interest at a rate of r per annum, compounding monthly. The corresponding effective annual rate of interest to this is 11.27%. **2**

Calculate the value of r as a percentage correct to two decimal places.

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Question 16 (3 marks)

Find the value of k such that $\displaystyle\int_{-5}^{-2} \frac{x^2}{x^3 - 2}\,dx = \ln k$. **3**

Question 17 (16 marks)

Let $f(x) = (x+2)(x-2)^3$.

(a) Write down the x-intercepts of $y = f(x)$. **1**

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(b) Show that $f'(x) = 4(x-2)^2(x+1)$ and $f''(x) = 12x(x-2)$. **2**

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(c) Find the coordinates of the stationary points of $y = f(x)$ and determine their nature. Justify your answers fully. **3**

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Question 17 continues on page 11

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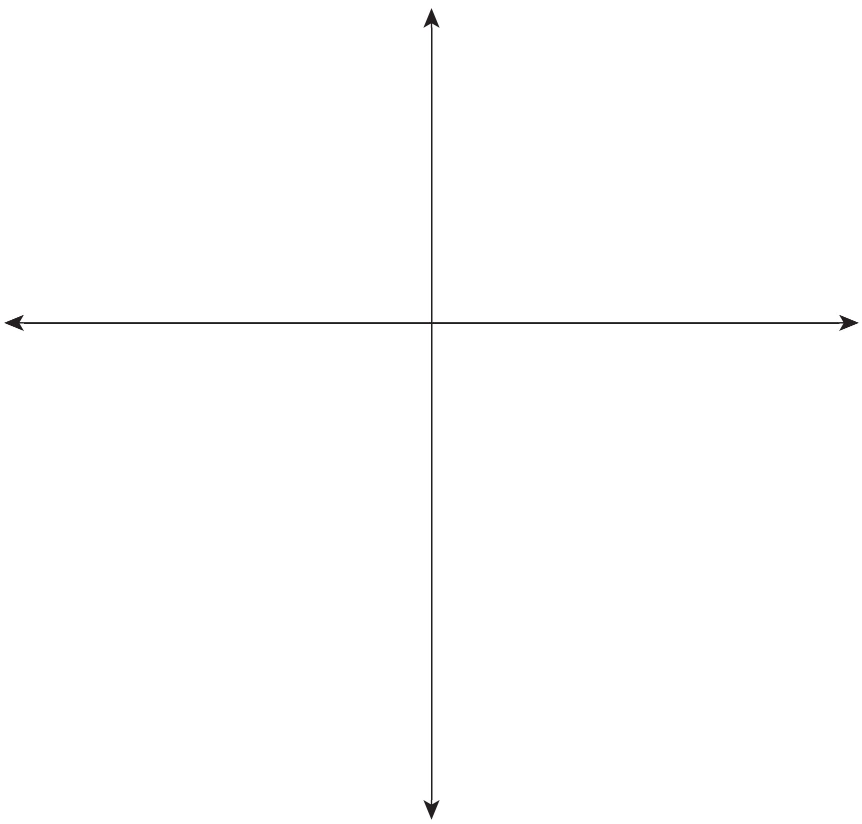
Question 17 (continued)

(d) Find the coordinates of all points of inflection of $y = f(x)$. **2**

(e) Sketch the graph of $y = f(x)$ on the axes below, showing all FOUR of the points found in parts (a), (b), and (c). **2**



Question 17 continues on page 12

Question 17 (continued)

(f) Does $y = f(x)$ have a global maximum or global minimum on its natural domain? If so, specify where. **1**

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(g) State, in the correct order, the transformations required to obtain the graph of $y = f\left(2\left(x - \frac{1}{4}\right)\right)$. **2**

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(h) On the set of axes provided in part (e), sketch the graph of $y = f\left(2\left(x - \frac{1}{4}\right)\right)$, showing its x-intercepts, stationary points and inflection points. **3**

End of Question 17

Question 18 (4 marks)

Consider the following extract from a table of FUTURE value interest factors, generated through the formula $A = P(1 + r)^n$.

n	*4%*	*8%*
0	1.0000	1.0000
1	1.0400	1.0800
2	1.0816	1.1664
3	1.1249	1.2587
4	1.1699	1.3605

(a) In an ordinary annuity, deposits are made at the end of every period. **2**

Calculate the PRESENT value of a 2 year ordinary annuity at rate 8% per annum, compounding every half year. Round your answer to the nearest cent.

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Question 18 continues on page 14

Question 18 (continued)

(b) A savings account is opened with deposits made at the end of each year. The interest rate is 8% per annum with annual compounding interest. Immediately after the fifth deposit, the amount in the account is \$1000.00. **2**

What is the amount of the annual contribution that must be made to achieve this? Round your answer to the nearest cent.

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End of Question 18

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Question 19 (4 marks)

(a) Find all solutions of $\tan\left(2x - \frac{\pi}{3}\right) = \frac{1}{\sqrt{3}}$ for all x in the interval $[0, 2\pi]$. **3**

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(b) The graph of $y = \tan\left(2x - \frac{\pi}{3}\right)$ is plotted along with its vertical asymptotes. **1**

Identify any ONE of the vertical asymptotes.

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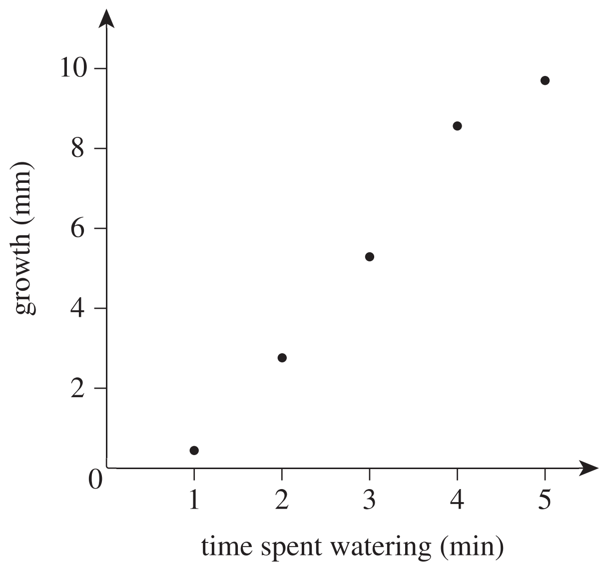
Question 20 (3 marks)

A study compared the growth of a plant in one day with the time spent watering the plant on that day. The measurements were taken on five separate days and are recorded in the scatter plot below.



The least squares regression line for this data is $y = -0.1263 + 2.0694x$. Do NOT prove this.

(a) A student attempts to use this model to determine how much their plant will grow in a day if they water it for 10 minutes on that day. **1**

How much growth does the model predict?

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(b) Is the prediction made in part (a) convincing? Justify your answer. **2**

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Question 21 (3 marks)

In an arithmetic sequence, the fifth term is 9 and the twenty-first term is 425. **3**

What is the smallest value of n such that the sum of the first n terms in the sequence is at least 10 487?

Question 22 (6 marks)

Alex opened up a new investment fund that gains interest at 6% per annum, compounding quarterly. He makes regular deposits of \$4000 at the end of every three months. The first deposit was made immediately after the account was opened.

Let A_n be the amount of money in Alex's fund after n quarter years, for $n \geq 0$.

(a) Show that if Alex maintains his deposits for nine full years, his fund will have reached a value of \$195 940.44 to the nearest cent. **3**

(b) What is the single sum required when invested under the same conditions that would reach this value after 9 years? Give your answer to the nearest cent. **1**

Question 22 continues on page 19

Question 22 (continued)

(c) After how many full months will Alex's fund reach \$500 000? **2**

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End of Question 22

Question 23 (10 marks)

The following table summarises the mean μ and standard deviation σ of a cohort's marks in English Advanced and Mathematics Advanced.

	Mean	*Standard deviation*
English Advanced mark	57	17
Mathematics Advanced mark	70	7

(a) Let X be the Mathematics Advanced mark of a randomly chosen student, prior to rounding. Suppose that X can be modelled by a normal distribution, with parameters as stated above. **1**

What is the value of $P(X > 70)$?

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(b) Assume that the probability density function of a normal random variable with mean μ and variance σ^2 is **4**

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Use the trapezoidal rule with FOUR sub-intervals to estimate $P(49 < X < 77)$ correct to four decimal places. Interpret your conclusions with reference to the empirical rule.

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Question 23 continues on page 21

Question 23 (continued)

(c) Let Y be the English Advanced mark of a randomly chosen student, prior to rounding. Suppose that Y can be modelled by a normal distribution in a similar way to X. **1**

The probability density functions of X and Y were plotted on the same set of axes. A student recorded the x-coordinate of each function's local maximum.

Which one of the random variables had the lesser value for the local maximum of its probability density function?

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(d) A student scored 76 in both English Advanced and Mathematics Advanced. **2**

Relative to their cohort, which subject did the student perform better in? Justify your answer with appropriate computations.

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(e) Another student attained the same mark for both English Advanced and Mathematics Advanced. She performed equally well relative to both cohorts. **2**

What mark has the student attained for both subjects?

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End of Question 23

Question 24 (7 marks)

The acceleration on a particle P_1 in m s^{-2} is $\dfrac{d^2x}{dt^2} = e^{-t} + e^{-2t}$ after t seconds. Initially, the particle is $\dfrac{3}{4}$ m to the right of the origin, travelling at velocity $\dfrac{dx}{dt} = -\dfrac{3}{2}$ m s^{-1}.

(a) Show that the displacement of the particle is given by $x = e^{-t} + \dfrac{1}{4}e^{-2t} - \dfrac{1}{2}$. **2**

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(b) Find the limiting displacement of P_1, and hence state the limiting distance that P_1 travels. **1**

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Question 24 continues on page 23

Question 24 (continued)

Another particle, P_2, moves simultaneously with the first particle. The acceleration P_2 experiences is

$$\frac{d^2x}{dt^2} = -e^{-t} - e^{-2t}.$$

This particle is $\frac{3}{4}$ m to the right of the origin and travelling at a velocity $\frac{dx}{dt} = -\frac{3}{2}$ m s^{-1} when $t = \ln 3$ seconds.

(c) Determine the exact time when P_1 and P_2 are travelling at the same velocity. Do NOT simplify your answer. **4**

End of Question 24

Question 25 (8 marks)

Consider the function below.

$$f(x) = \begin{cases} \frac{1}{2}x & \text{if } 0 \le x \le 1, \\ \frac{2}{3} - \frac{1}{6}x & \text{if } 1 < x \le 4, \\ 0 & \text{else} \end{cases}$$

You may assume that $f(x) \ge 0$ for all x in $(-\infty, \infty)$.

(a) Prove the other property required to show that $f(x)$ is a valid density function for a continuous random variable X. **2**

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(b) Sketch $y = f(x)$, showing each point where the rule defining $y = f(x)$ changes. **2**

Question 25 continues on page 25

Question 25 (continued)

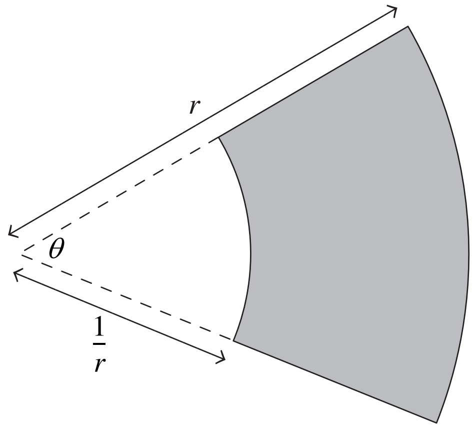
(c) State the mode of X. **1**

(d) Find the median of X. **3**

End of Question 25

Question 26 (5 marks)

An annulus sector is made at angle θ such that the inner and outer radii are r metres and $\frac{1}{r}$ metres respectively, as shown in the diagram.



Assume that the perimeter of this annulus sector is 6 metres.

(a) Show that $\theta = \dfrac{2(-r^2 + 3r + 1)}{r^2 + 1}$. **2**

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Question 26 continues on page 27

(b) Assume without proof that $-2r^4 + 3r^3 + 3r + 2 = -(2r + 1)(r - 2)(r^2 + 1)$. **3**

If the area of the annulus sector is $A = -r^2 + 3r + 2 - \frac{3}{r} - \frac{1}{r^2}$, find the maximum possible area of the annulus.

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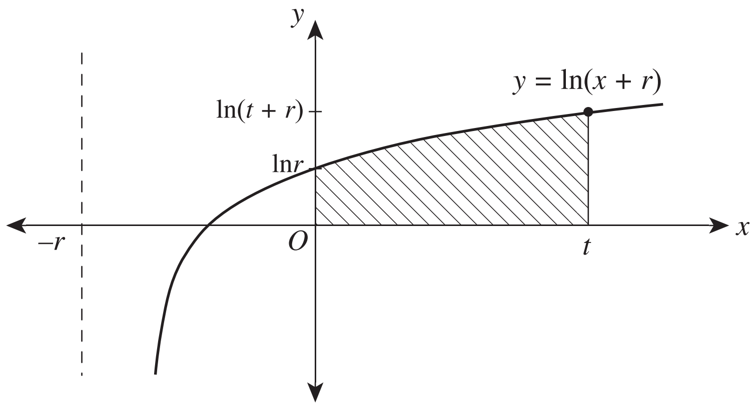
End of Question 26

Question 27 (3 marks)

Consider the graph of $y = \ln(x + r)$ shown below. Let $r > 1$. **3**



The region bound by the curve $y = \ln(t + r)$, the coordinate axes, and the line $x = t$ is used as a model for a garden, where $t > 0$.

You may assume without proof that the equation of the curve is equivalently expressed as $x = e^y - r$.

Show that the area of this region, A, in terms of r and t is $A = (t + r)\ln(t + r) - r\ln r - t$.

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Question 28 (5 marks)

It is known that the angles in a triangle sum to π. In a particular triangle, the angles form a geometric progression. One of the angles is $\frac{\pi}{4}$. **5**

Find the TWO possible configurations of the angles of this triangle. Give your answers in simplified, exact form.

End of paper

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi r h$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1},\ r \neq 1$$

$$S = \frac{a}{1 - r},\ |r| < 1$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

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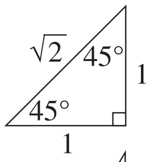
Trigonometric Functions

$\sin A = \dfrac{\text{opp}}{\text{hyp}}, \quad \cos A = \dfrac{\text{adj}}{\text{hyp}}, \quad \tan A = \dfrac{\text{opp}}{\text{adj}}$

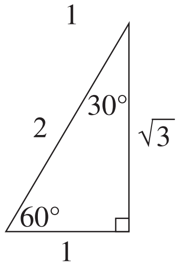
$A = \dfrac{1}{2}ab\sin C$

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$



$c^2 = a^2 + b^2 - 2ab\cos C$

$\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$



$l = r\theta$

$A = \dfrac{1}{2}r^2\theta$

Trigonometric identities

$\sec A = \dfrac{1}{\cos A}, \cos A \neq 0$

$\operatorname{cosec} A = \dfrac{1}{\sin A}, \sin A \neq 0$

$\cot A = \dfrac{\cos A}{\sin A}, \sin A \neq 0$

$\cos^2 x + \sin^2 x = 1$

Compound angles

$\sin(A + B) = \sin A\cos B + \cos A\sin B$

$\cos(A + B) = \cos A\cos B - \sin A\sin B$

$\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A\tan B}$

If $t = \tan\dfrac{A}{2}$ then $\sin A = \dfrac{2t}{1 + t^2}$

$\cos A = \dfrac{1 - t^2}{1 + t^2}$

$\tan A = \dfrac{2t}{1 - t^2}$

$\cos A\cos B = \dfrac{1}{2}[\cos(A - B) + \cos(A + B)]$

$\sin A\sin B = \dfrac{1}{2}[\cos(A - B) - \cos(A + B)]$

$\sin A\cos B = \dfrac{1}{2}[\sin(A + B) + \sin(A - B)]$

$\cos A\sin B = \dfrac{1}{2}[\sin(A + B) - \sin(A - B)]$

$\sin^2 nx = \dfrac{1}{2}(1 - \cos 2nx)$

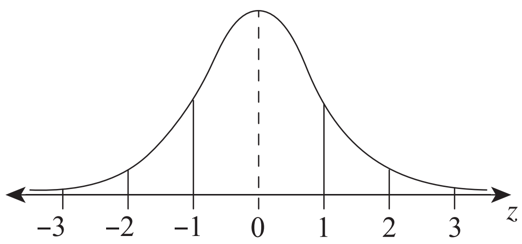
$\cos^2 nx = \dfrac{1}{2}(1 + \cos 2nx)$

Statistical Analysis

$z = \dfrac{x - \mu}{\sigma}$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

$E(X) = \mu$

$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$

Probability

$P(A \cap B) = P(A)P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A|B) = \dfrac{P(A \cap B)}{P(B)}, P(B) \neq 0$

Continuous random variables

$P(X \leq r) = \displaystyle\int_a^r f(x)dx$

$P(a < X < b) = \displaystyle\int_a^b f(x)dx$

Binomial distribution

$P(X = r) = {}^nC_r p^r(1 - p)^{n - r}$

$X \sim \operatorname{Bin}(n, p)$

$\Rightarrow P(X = x)$

$= \dbinom{n}{x} p^x(1 - p)^{n - x}, x = 0, 1, \ldots, n$

$E(X) = np$

$\operatorname{Var}(X) = np(1 - p)$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\dfrac{dy}{dx} = nf'(x)[f(x)]^{n-1}$
$y = uv$	$\dfrac{dy}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$
$y = \dfrac{u}{v}$	$\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\dfrac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\dfrac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\dfrac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\dfrac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\dfrac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{\sqrt{1 - [f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\dfrac{dy}{dx} = -\dfrac{f'(x)}{\sqrt{1 - [f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{1 + [f(x)]^2}$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)dx$$

$$\approx \frac{b-a}{2n}\{f(a) + f(b) + 2[f(x_1) + \ldots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

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Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \ldots + \binom{n}{r}x^{n-r}a^r + \ldots + a^n$$

Vectors

$$|\underset{\sim}{u}| = |x\underset{\sim}{i} + y\underset{\sim}{j}| = \sqrt{x^2 + y^2}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = |\underset{\sim}{u}||\underset{\sim}{v}|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta)$$
$$= re^{i\theta}$$

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$$
$$= r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$